SOFI

Smart Accessibility Companion

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Detailed Project Proposal

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Bilingual (English / Urdu) | Voice-First | PWA

Confidential — For Review Purposes Only

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1. Executive Summary

SOFI (Smart Accessibility Companion) is a bilingual, voice-first Progressive Web Application designed to empower individuals with disabilities — including visual, hearing, motor, speech, and cognitive impairments — through an intelligent, personalized digital assistant.

The application combines real-time voice recognition, AI-powered natural language understanding, and adaptive user interfaces to provide a comprehensive accessibility platform that serves as a daily companion for users who face barriers interacting with conventional technology.

Key differentiators:

- Voice-first architecture with always-on speech recognition and fuzzy matching
- Full bilingual support (English and Urdu) including Urdu speech synthesis
- Adaptive UI that reconfigures based on the user's specific disability type
- AI companion powered by Google Gemini for conversational assistance
- Emergency SOS system with GPS location sharing and automated contact alerts
- Installable PWA — no app store required, works on any smartphone browser

2. Problem Statement

According to the World Health Organization, over 1.3 billion people globally live with some form of disability. In Pakistan alone, the Pakistan Bureau of Statistics reports approximately 3.3 million persons with disabilities, though actual numbers are estimated to be significantly higher.

These individuals face critical barriers in daily life:

- Mainstream apps are designed for able-bodied users, ignoring accessibility needs
- Voice assistants like Siri and Google Assistant lack Urdu support and disability-specific features
- Emergency communication tools don't account for users who cannot speak, see, or type quickly
- Health tracking, medication reminders, and navigation require separate apps with separate learning curves
- Cognitive disabilities are almost entirely overlooked in app design

There is no unified, bilingual, accessibility-first platform that addresses all these needs in a single, simple interface.

3. Proposed Solution

SOFI addresses these gaps by providing an all-in-one accessibility companion that:

1. Detects and adapts to the user's disability type during onboarding
2. Offers voice-first interaction as the primary input method with text fallback
3. Provides an AI companion that understands context and assists with daily tasks
4. Integrates emergency, health, navigation, communication, and productivity in one app
5. Supports both English and Urdu at every level — UI, voice, and AI responses
6. Deploys as a lightweight PWA, eliminating app store barriers

4. Target Audience

User Segment	Needs	SOFI Solution
Visually Impaired	Voice-only navigation, screen reader support	Always-on voice recognition, spoken confirmations, voice-first mode
Hearing Impaired	Visual cues, text-based interaction	Text command input, visual indicators, high-contrast UI
Motor Disabilities	Large touch targets, minimal gestures	44px minimum touch areas, voice control, simplified layouts
Speech Impaired	Text-based commands, quick actions	Text command bar, pre-built quick actions, keyboard navigation
Cognitive Disabilities	Simplified interface, fewer choices	Reduced feature set, larger elements, step-by-step guidance
Elderly Users	Easy to learn, reliable emergency access	One-tap SOS, large fonts, voice-guided tutorial
Caregivers	Monitor & assist remotely	Activity logs, emergency notifications, health dashboard

5. Core Features & Modules

5.1 AI Companion (SOFI Chat)

- Conversational AI powered by Google Gemini 3 Flash Preview
- Accepts both voice and text input seamlessly
- Responds in the user's preferred language (English or Urdu)
- Contextual awareness — remembers conversation history within a session
- Markdown-rendered responses for rich formatting

5.2 Voice Command System

- Always-on SpeechRecognition API with continuous listening
- Fuzzy keyword matching using Dice coefficient similarity (threshold: 0.6-0.75)
- Intent detection: navigate, call, message, reminder, emergency, unknown
- Confidence filtering with configurable minimum threshold (0.4)
- Real-time interim transcript display for user feedback
- Page-specific voice handlers for contextual commands

5.3 Emergency SOS

- One-tap SOS button with 5-second countdown (cancel window)
- Voice-activated: say "Emergency" or "SOS" anywhere in the app
- GPS location detection and sharing with emergency contacts
- Sequential notification of up to 5 emergency contacts
- Activity logging for post-incident review
- Voice confirmation before activation to prevent accidental triggers

5.4 Contacts & Communication

- Contact management with voice-triggered calling
- SMS/message composition with voice or text input
- Fuzzy name matching for voice commands ("Call Ahmed")
- Emergency contact designation

5.5 Health Dashboard

- Track heart rate, blood pressure, steps, sleep, mood, and pain level
- Medication tracking with reminders
- Medical profile: blood type, allergies, conditions, medications
- Visual charts powered by Recharts

5.6 Reminders & Notes

- Voice-created reminders with title, time, and recurrence
- Recurring options: once, daily, weekly, monthly
- Voice-to-text note capture
- Bilingual titles (English + Urdu)

5.7 Navigation

- Interactive maps powered by Leaflet.js
- Turn-by-turn routing via Leaflet Routing Machine
- Destination history saved per user
- Voice command: "Navigate to [place]"

5.8 Settings & Personalization

- Disability type selection with UI adaptation
- Font size: normal, large, extra-large
- High-contrast mode toggle
- Voice speed: slow, normal, fast
- Language toggle: English ↔ Urdu
- Voice diagnostics panel for debugging
- Tutorial replay functionality

6. Technical Architecture

6.1 Frontend Stack

Technology	Purpose
React 18	Component-based UI with hooks and context API
TypeScript 5	Type safety, better developer experience, fewer runtime errors
Vite 5	Fast build tool with HMR, optimized production builds
Tailwind CSS 3	Utility-first CSS with semantic design tokens
Framer Motion	Smooth animations and gesture support
shadcn/ui	Accessible, customizable component library (Radix primitives)
React Router 6	Client-side routing with protected routes
TanStack Query	Server state management, caching, and synchronization
Leaflet.js	Interactive maps and routing
Recharts	Health metric data visualization

6.2 Backend (Lovable Cloud)

- PostgreSQL database with Row-Level Security (RLS)
- Authentication with email/password and email verification
- Edge Functions for AI chat processing (Google Gemini integration)
- Real-time subscriptions for live data updates
- File storage for user assets

6.3 Web APIs

- Web Speech API — SpeechRecognition (voice input) and SpeechSynthesis (TTS)
- Geolocation API — GPS coordinates for emergency and navigation
- Permissions API — Microphone permission status detection
- Vibration API — Haptic feedback for native mobile feel
- Web App Manifest — PWA installability and standalone display

7. Accessibility Framework

SOFI implements a multi-layered accessibility system that adapts the entire application based on the user's disability profile:

7.1 Disability-Specific Adaptations

Disability	UI Adaptation	Interaction Model
Visual	Minimal visual elements, high contrast	Voice-only mode, spoken confirmations for every action
Hearing	Always-visible text input, visual alerts	Text commands, no audio required
Motor	Extra-large touch targets (44px+)	Voice commands + large buttons
Cognitive	Reduced features (4 core only), larger cards	Step-by-step guidance, simplified language
Speech	Prominent text input bar	Text-based commands, quick action buttons

7.2 WCAG 2.1 Compliance

- Semantic HTML with ARIA labels, roles, and live regions
- Keyboard navigable interface
- High contrast mode with configurable color scheme
- Scalable font sizes (16px, 20px, 24px)
- Focus indicators and skip navigation
- RTL text direction support for Urdu

8. Database Schema

The application uses 8 core tables with Row-Level Security policies ensuring data isolation per user:

Table	Purpose & Key Fields
profiles	User preferences: disability type, language, font size, voice settings, medical info
emergency_contacts	SOS contacts: name, phone, relationship (max 5 per user)
chat_messages	AI conversation history: role, content, bilingual content, message type
reminders	Scheduled reminders: title (EN/UR), time, recurrence, completion status
notes	Text/voice notes: title, content, timestamps
health_metrics	Health data: metric type, value, unit, recording timestamp
navigation_history	Saved destinations: name, address, coordinates
activity_logs	Audit trail: action, category, metadata (JSON), timestamp

9. AI & Voice Intelligence

9.1 Voice Recognition Pipeline

The voice system operates as a continuous pipeline:

1. Continuous Listening: Web SpeechRecognition API runs in always-on mode with auto-restart on silence (500ms delay)
2. Interim Display: Real-time transcript shown to user as they speak
3. Confidence Filtering: Results below 0.4 confidence are rejected; 0 confidence (unreported) is accepted
4. Multi-Alternative Selection: Up to 3 alternative transcriptions compared, best selected
5. Intent Parsing: Fuzzy keyword matching with Dice coefficient (similarity threshold 0.6-0.75)
6. Action Execution: Navigate, call, message, create reminder, trigger emergency, or delegate to AI

9.2 AI Chat Integration

- Model: Google Gemini 3 Flash Preview (via Lovable AI Gateway)
- No API key required — integrated through platform services
- Maintains conversation context within session
- Handles unknown voice intents as conversational queries
- Markdown-formatted responses with bilingual support

10. Security & Privacy

- Row-Level Security (RLS) on all database tables — users can only access their own data
- Authentication via email/password with email verification
- JWT token-based session management
- No client-side storage of sensitive credentials
- Edge functions run server-side with secure environment variables
- HTTPS-only communication
- Microphone permission requested explicitly with clear user explanation
- Activity logging for audit trail (accessible to user and authorized caregivers)
- Medical data stored encrypted at rest in PostgreSQL

11. Deployment Strategy (PWA)

SOFI is deployed as a Progressive Web App (PWA), providing near-native mobile experience without app store friction:

- Web App Manifest with standalone display mode for full-screen experience
- Installable on Android (Add to Home Screen prompt) and iOS (manual Add to Home Screen)
- Safe area insets for notched devices (iPhone X+, modern Android)
- Touch-optimized: disabled overscroll bounce, tap highlight, and text selection on interactive elements
- Mobile-first responsive design with 320px minimum viewport support
- Platform detection for tailored installation instructions
- Published via Lovable's hosting with custom domain support

12. Development Timeline

Phase	Deliverables	Duration	Status
Phase 1	Core architecture, auth, profile setup, accessibility framework	3 weeks	Completed
Phase 2	Voice system, AI companion, emergency SOS, contacts	4 weeks	Completed
Phase 3	Health dashboard, reminders, notes, navigation, messages	4 weeks	Completed
Phase 4	PWA optimization, voice diagnostics, tutorial, mobile polish	2 weeks	Completed
Phase 5	User testing, caregiver portal, push notifications, offline mode	4 weeks	Planned
Phase 6	Community feedback, localization (Sindhi, Pashto), app store wrapper	6 weeks	Future

Total estimated timeline: 23 weeks (Phases 1-4 complete, Phase 5-6 pending)

13. Budget Estimate

Item	Monthly Cost	Annual Cost
Lovable Cloud (Backend + Hosting)	$25 - $50	$300 - $600
AI Gateway (Gemini API calls)	$10 - $30	$120 - $360
Custom Domain	$1 - $2	$12 - $24
Map Tiles (if premium)	$0 - $15	$0 - $180
SMS/Notification Service (Phase 5)	$5 - $20	$60 - $240
Total Estimated	**$41 - $117**	**$492 - $1,404**

Note: Development costs are not included as the project is built on the Lovable platform. The above represents operational costs only.

14. Risk Assessment & Mitigation

Risk	Severity	Likelihood	Mitigation
Browser speech API inconsistency	High	Medium	Fallback text input, error recovery with auto-restart, diagnostics panel
Urdu speech recognition accuracy	Medium	High	Fuzzy matching, multiple alternatives, confidence display for debugging
Microphone permission denied	High	Medium	Graceful degradation to text mode, clear error messages, permission guide
Data privacy concerns	High	Low	RLS policies, no third-party data sharing, transparent logging
PWA limitations on iOS	Medium	High	Platform-specific install instructions, Capacitor wrapper as backup
AI response quality	Medium	Low	Using top-tier Gemini model, system prompts optimized for accessibility context

15. Future Roadmap

Near-term (3-6 months)

- Push notifications for reminders and emergency alerts
- Offline mode with service worker caching
- Caregiver portal for remote monitoring
- WhatsApp integration for messaging
- Real-time medication interaction checking

Mid-term (6-12 months)

- Additional language support: Sindhi, Pashto, Punjabi
- Wearable device integration (smartwatch companion)
- Computer vision module for object/text recognition via camera
- Community features: forums, peer support groups
- Government accessibility compliance certification

Long-term (12-24 months)

- Sign language recognition via camera AI
- Predictive health analytics with ML models
- Integration with national health systems
- Multi-platform native apps via Capacitor
- Partnerships with disability organizations and NGOs

16. Conclusion

SOFI represents a paradigm shift in how technology serves persons with disabilities, particularly in bilingual communities. By combining voice-first interaction, adaptive UI, AI-powered assistance, and emergency capabilities in a single, lightweight PWA, SOFI removes the barriers that have traditionally excluded millions from benefiting from digital technology.

The platform is already functional with core features complete (Phases 1-4), demonstrating the viability of the approach. With continued development focused on user testing, caregiver features, and expanded language support, SOFI has the potential to become the definitive accessibility companion for Urdu-speaking communities and beyond.

We invite stakeholders, accessibility advocates, and potential partners to review this proposal and join us in making technology truly accessible for everyone.

— End of Proposal —